SST Falsifier Report

A054-v0.5.1
E013 Provider-Convergent State Selection (source-ledger reconstruction)
External read-only rendering; not an original frozen report.
Generated: 2026-10-09T17:35:40.189911+00:00

Evidence boundary

No new science was executed for this report. Gate outcomes are copied from the source ledger without reclassification. Missing records are not silently filled.

Gate summary

DEFERRED=1, PASS=6, UNRESOLVED=7

Gate	Status	Preregistered question
G0	PASS	Is the frozen v0.5.1 protocol intact and runtime/implementation provenance self-consistent?
G1	PASS	Is the run exactly bound to the E013 common-carrier manifest, with all E011 source geometries reloaded and hash-verified?
G2	PASS	Do rigid-objectivity, constrained-objective identity, Armijo monotonicity and topology-proxy controls pass?
G3	UNRESOLVED	Can the fixed target-free constrained-objective selector drive the preregistered carrier fraction to stationary states, with numerical descent failures retained as UNRESOLVED?
G4	UNRESOLVED	Does the same frozen constrained-objective selector generalize without refit to registered confirmation topologies?
G5	PASS	Does C++/OpenMP FP64 reproduce Python/NumPy FP64 for the state-selection energy kernel?
G6	PASS	If available, does framework GPU/DD32 screening evidence remain consistent without being promoted to certification?
G7	UNRESOLVED	Do independently sourced provider anchors of the same topology converge to the same stationary endpoint without unresolved optimizer cases?
G8	UNRESOLVED	Are stationarity-qualified endpoints locally stable under a finite-difference-converged constrained Hessian and bounded ringdown on the same objective?
G10	UNRESOLVED	For source-energy-sensitive topologies, does numerically resolved state selection collapse provider spread toward a common endpoint?
G11	PASS	Are clearance and link-drift topology proxies preserved along every completed state-selection path?
G12	UNRESOLVED	Can at least one provider-converged stationary, Hessian-converged, stable state be frozen as an A057-compatible handoff?
G22	UNRESOLVED	What is the overall state-selection evidence classification when numerical unresolved states are kept distinct from scientific FAIL?
G9	DEFERRED	After a completed FULL/CERTIFY blind run, do reveal and commitment checks verify exactly?

Gate detail (source-derived)

G0 — PASS

Is the frozen v0.5.1 protocol intact and runtime/implementation provenance self-consistent?

Framework protocol and implementation commitment verified before science.

- diagnostic_continuation: True
- mode: BASIC

G1 — PASS

Is the run exactly bound to the E013 common-carrier manifest, with all E011 source geometries reloaded and hash-verified?

Exact E013/E011 selection and source-native bytes are required; no legacy substitute is permitted.

- admitted: 5
- e013_manifest_sha256: e0e2107e3ffcf1117829da0071afcb6dae5384f150921081ccb725427b86f777
- geometry_hashes_verified: True
- rejected: 0
- unique_static_seed_ids: True

G2 — PASS

Do rigid-objectivity, constrained-objective identity, Armijo monotonicity and topology-proxy controls pass?

Synthetic controls validate objectivity and verify that derivatives, acceptance and endpoint diagnostics use the same constrained objective.

- circle_energy_nonincrease: True
- circle_topology_proxy_ok: True
- constrained_objective_consistency: True
- rigid_energy_rel_l2: 3.480951510493397e-16
- selector_numerical_status: RESOLVED

G3 — UNRESOLVED

Can the fixed target-free constrained-objective selector drive the preregistered carrier fraction to stationary states, with numerical descent failures retained as UNRESOLVED?

A numerical failure to demonstrate descent is retained as UNRESOLVED_NUMERICAL rather than counted as scientific evidence against state selection.

- completed_carriers: 5
- median_stationarity_reduction_fraction: -0.025717206662263514
- numerically_unresolved_carriers: 4
- required_fraction: 0.5
- stationary_topology_preserved_carriers: 1

G4 — UNRESOLVED

Does the same frozen constrained-objective selector generalize without refit to registered confirmation topologies?

Selector parameters are frozen globally and are not refit on confirmation topologies; numerical indeterminacy is not promoted to a scientific FAIL.

- confirmation_carriers: 1
- confirmation_topologies: array, 1 elements
- numerically_unresolved: 1
- required_fraction: 0.5
- success_fraction_lower_bound: 0.0

G5 — PASS

Does C++/OpenMP FP64 reproduce Python/NumPy FP64 for the state-selection energy kernel?

C++/OpenMP FP64 is certification authority only for the shared energy kernel.

- relative_l2: 0.0
- threshold: 1e-10

G6 — PASS

If available, does framework GPU/DD32 screening evidence remain consistent without being promoted to certification?

GPU/DD32 remains screening-only.

- screening_evidence_files: array, 1 elements

G7 — UNRESOLVED

Do independently sourced provider anchors of the same topology converge to the same stationary endpoint without unresolved optimizer cases?

BASIC may contain only one provider anchor per topology.

- eligible_topologies: 0

G8 — UNRESOLVED

Are stationarity-qualified endpoints locally stable under a finite-difference-converged constrained Hessian and bounded ringdown on the same objective?

Stationarity is necessary but not sufficient; Hessian sign is promoted only after eps/2/4 finite-difference convergence, then bounded nonlinear ringdown is required.

- hessian_numerically_unresolved_endpoints: 1
- required_fraction: 0.5
- stable_fraction_lower_bound: 0.0
- stable_fraction_upper_bound: 1.0
- stable_ringdown_bounded_endpoints: 0

G10 — UNRESOLVED

For source-energy-sensitive topologies, does numerically resolved state selection collapse provider spread toward a common endpoint?

No replicated topology exceeds the preregistered source-energy sensitivity threshold.

- source_sensitive_topologies: 0

G11 — PASS

Are clearance and link-drift topology proxies preserved along every completed state-selection path?

Continuous small-step evolution plus clearance and linking-drift checks is a topology-preservation proxy, not a complete knot-polynomial proof.

- completed_paths: 5
- topology_proxy_preserved_paths: 5
- violations: 0

G12 — UNRESOLVED

Can at least one provider-converged stationary, Hessian-converged, stable state be frozen as an A057-compatible handoff?

Only provider-converged, stationary, Hessian-converged, locally stable and ringdown-bounded states are promoted to a future A057 state-selected handoff.

- eligible_topologies: array, 0 elements
- eligible_topology_count: 0
- handoff_entries: 0
- upstream_numerically_or_evidentially_unresolved: True

G22 — UNRESOLVED

What is the overall state-selection evidence classification when numerical unresolved states are kept distinct from scientific FAIL?

Overall classification is blind to mass targets and old falsifier results.

- classification: STATE_SELECTION_UNRESOLVED_AT_REGISTERED_LEVEL

G9 — DEFERRED

After a completed FULL/CERTIFY blind run, do reveal and commitment checks verify exactly?

Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Backend provenance

{"cpp_certification": {"authority": "CERTIFICATION", "gate_status": "PASS"}, "diagnostic_continuation": true, "gpu_screening": {"authority": "SCREENING_ONLY", "gate_status": "PASS"}, "python_reference": {"actual_backend": "python-numpy-fp64", "authority": "REFERENCE", "precision": "float64"}, "schema": "SST-BACKEND-MANIFEST-2", "scientific_gate_dependencies": "soft; see DEPENDENCY_CAVEATS.json"}

Report provenance

Declared ledger-to-summary digest binding: True
Files read: BACKEND_MANIFEST.json, DEPENDENCY_CAVEATS.json, GATE_LEDGER.json, RUN_SUMMARY.json
Scope: Read-only digest inventory and declared ledger/summary binding check only. Does not independently certify framework protocol, scientific results, or frozen output tree.